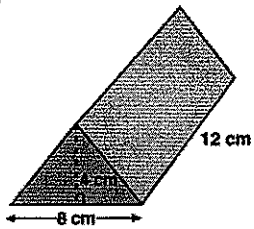


8.3 Volume of Prisms and Pyramids

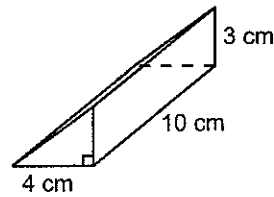
Section 1: Find the Volume of the following prisms

1)



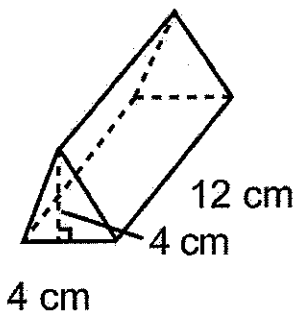
$$\begin{aligned} V &= \frac{1}{2} b l h \\ &= \frac{1}{2} (8)(4)(12) \\ &= 192 \text{ cm}^3 \end{aligned}$$

2)



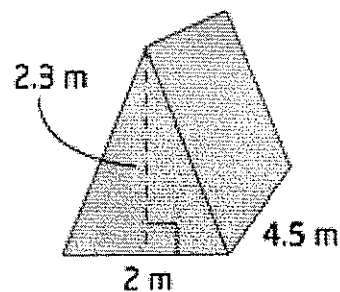
$$\begin{aligned} V &= \frac{1}{2} b l h \\ &= \frac{1}{2} (4)(3)(10) \\ &= 60 \text{ cm}^3 \end{aligned}$$

3)



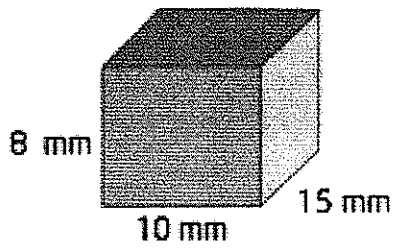
$$\begin{aligned} V &= \frac{1}{2} b l h \\ &= \frac{1}{2} (4)(4)(12) \\ &= 96 \text{ cm}^3 \end{aligned}$$

4)



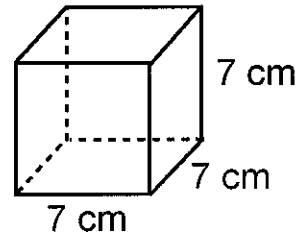
$$\begin{aligned} V &= \frac{1}{2} b l h \\ &= \frac{1}{2} (2)(2.3)(4.5) \\ &= 10.35 \text{ m}^3 \end{aligned}$$

5)



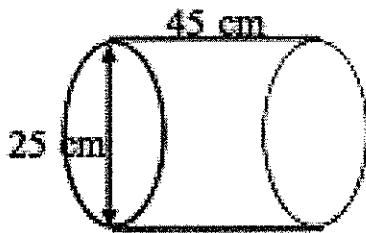
$$\begin{aligned} V &= lwh \\ &= (15)(10)(8) \\ &= 1200 \text{ mm}^3 \end{aligned}$$

6) Express your answer in Litres



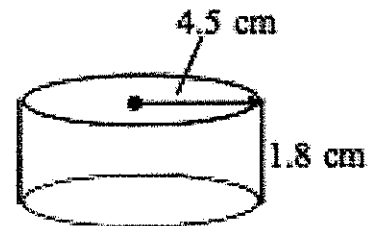
$$\begin{aligned} V &= s^3 \\ V &= (7)^3 \\ V &= 343 \text{ cm}^3 \end{aligned}$$

7)



$$\begin{aligned} V &= \pi r^2 h \\ V &= \pi (12.5)^2 (45) \\ &= 22089.3 \text{ cm}^3 \end{aligned}$$

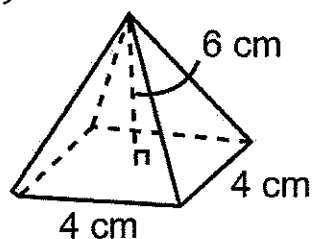
8)



$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (4.5)^2 (1.8) \\ &= 114.5 \text{ cm}^3 \end{aligned}$$

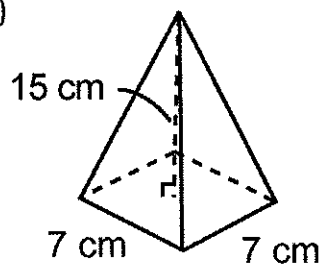
Section 2: Find the Volume of the following pyramids

9)



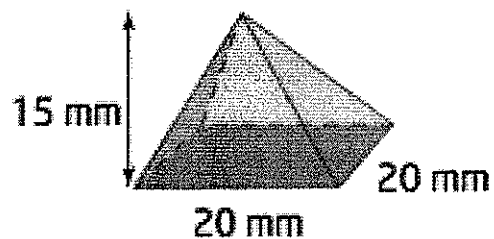
$$V = \frac{1}{3}(4^2)(6)$$
$$= 32 \text{ cm}^3$$

10)



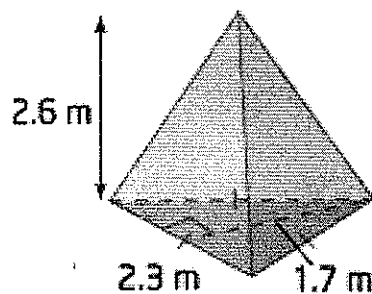
$$V = \frac{1}{3}(7^2)(15)$$
$$= 245 \text{ cm}^3$$

11)



$$V = \frac{1}{3}(20^2)(15)$$
$$= 2000 \text{ mm}^3$$

12)



$$V = \frac{1}{3}\left(\frac{1}{2}\right)(2.3)(1.7)(2.6)$$
$$= 1.7 \text{ m}^3$$

Section 3: Applications

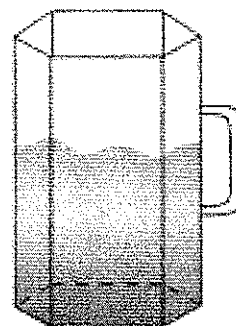
13) Sneferu's North Pyramid at Dahshur, Egypt has a square base with side length 220 meters and its height is 105 meters. What is the volume of the famous pyramid?

$$V = \frac{1}{3}(220^2)(105) \\ = 1\,694\,000\text{ m}^3$$

14) The picture of milk shown is a prism. The base has an area of 40 cm^2 and the height of the pitcher is 26 cm. Will the pitcher hold 1 Liter of milk?

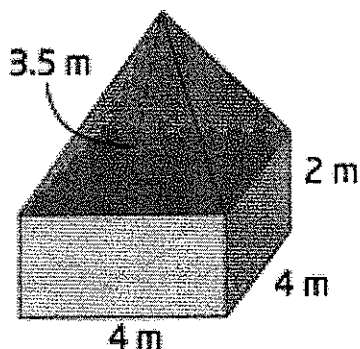
$$V = (40)(26) \\ = 1040\text{ cm}^3$$

$$1040\text{ cm}^3 \times \frac{1\text{ L}}{1000\text{ cm}^3} = 1.04\text{ L}$$



Yes it will hold 1 L of milk.

15) Find the volume of the following garden shed:



$$V_{\text{rectangular prism}} = lwh \\ = (4)(4)(2) \\ = 32\text{ m}^3$$

$$V_{\text{pyramid}} = \frac{1}{3}(4^2)(2.9) \\ = 15.5\text{ m}^3$$

Height of Pyramid:

$$h^2 = 3.5^2 - 2^2$$

$$h = 2.9\text{ m}$$

$$\text{Volume of Shed} = 32 + 15.5 \\ = 47.5\text{ m}^3$$